



Introduction to CFD (4RC30)

**Convection-diffusion problems;
Discretization schemes**

Prof.dr.ir. Niels Deen, N.G.Deen@tue.nl, Tel. 3681, VEC 3.202

Dr. Yali Tang, y.tang2@tue.nl, Tel. 8052, VEC 3.106

Summary of last lecture

- Rate of change + convection = diffusion + (source – sink)

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- General form of discretized equations:

$$\boxed{a_P \phi_P = \sum a_{nb} \phi_{nb} + S_u} \quad \boxed{a_P = \sum a_{nb} - S_P}$$

- Source terms are included through S_u and S_P

Content of this lecture (chapter 5)

- Discretization 1D steady convection-diffusion equation
- Discretization schemes and their properties
 - Central differencing scheme
 - Upwind differencing scheme
 - Hybrid differencing scheme
 - Other schemes

Learning outcomes of this session

1. **Formulate** a conservation equation for convective transport.
2. **Explain** the central, upwind, and hybrid discretization schemes using own words,
3. **Motivate** the choice for a discretization scheme for convective transport *for a giving physical problem*.

Steady convection-diffusion

- convection = diffusion + (source – sink) $\frac{\cancel{\partial(\rho\phi)}}{\cancel{\partial t}} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$

$$\text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- Integration over control volume: $\int_{CV} \text{div } \mathbf{a} dV = \int_A \mathbf{n} \cdot \mathbf{a} dA$

$$\int_A \mathbf{n} \cdot (\rho\phi\mathbf{u}) dA = \int_A \mathbf{n} \cdot (\Gamma \text{grad } \phi) dA + \int_{CV} S_\phi dV$$

- Assumptions:
 - $\mathbf{u}$ is known (computation of $\mathbf{u}$ is explained in lecture 3)
 - No sources or sinks

Steady 1D convection-diffusion

- **Transport equation:**

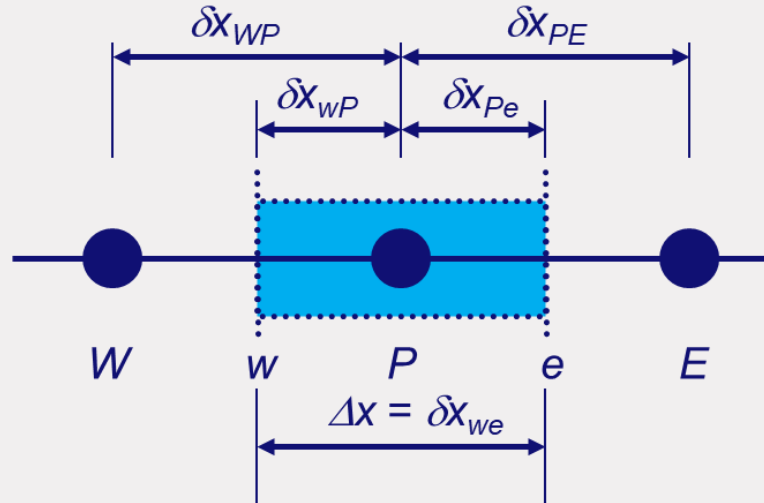
$$\frac{d}{dx}(\rho u \phi) = \frac{d}{dx} \left(\Gamma \frac{d\phi}{dx} \right)$$

- **Continuity equation (mass balance):**

$$\frac{d(\rho u)}{dx} = 0$$

Steady 1D convection-diffusion

Step 1: Grid generation



Steady 1D convection-diffusion

Step 2: Integration

- Transport equation: $\frac{d}{dx}(\rho u \phi) = \frac{d}{dx} \left(\Gamma \frac{d\phi}{dx} \right)$

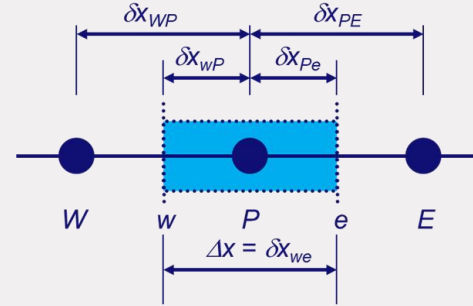
$$\int_A \mathbf{n} \cdot (\rho \phi \mathbf{u}) dA = \int_A \mathbf{n} \cdot (\Gamma \text{grad } \phi) dA$$

$$(\rho u A \phi)_e - (\rho u A \phi)_w = \left(\Gamma A \frac{d\phi}{dx} \right)_e - \left(\Gamma A \frac{d\phi}{dx} \right)_w$$

- Continuity equation (mass balance):

$$\frac{d(\rho u)}{dx} = 0$$

$$(\rho u A)_e - (\rho u A)_w = 0$$



Definition of F , D and Pe

- We define the convective mass flux per unit area F : $F = \rho u$
the diffusion conductance D : $D = \frac{\Gamma}{\delta x}$
the Peclet number Pe : $Pe = F / D = \frac{\rho u}{\Gamma / \delta x}$
- At the cell faces e and w :

$$F_w = (\rho u)_w \quad F_e = (\rho u)_e$$

$$D_w = \frac{\Gamma_w}{\delta x_{WP}} \quad D_e = \frac{\Gamma_e}{\delta x_{PE}}$$

Steady 1D convection-diffusion

- Integrated convection-diffusion equation:

$$(\rho u A \phi)_e - (\rho u A \phi)_w = \left(\Gamma A \frac{d\phi}{dx} \right)_e - \left(\Gamma A \frac{d\phi}{dx} \right)_w$$

$\left(\Gamma A \frac{d\phi}{dx} \right)_e = \Gamma_e A_e \left(\frac{\phi_E - \phi_P}{\delta x_{PE}} \right)$

Assuming $A_e = A_w = A$


$$F_e \phi_e - F_w \phi_w = D_e (\phi_E - \phi_P) - D_w (\phi_P - \phi_W)$$

- Integrated continuity equation:

$$(\rho u A)_e - (\rho u A)_w = 0 \quad \longrightarrow \quad F_e - F_w = 0$$

Assuming u is known

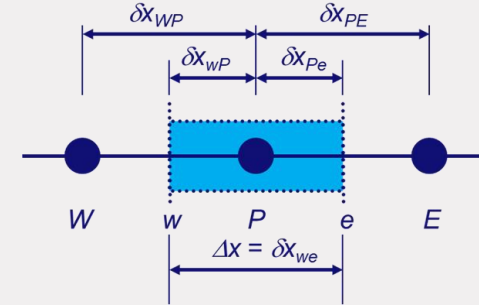
- We need a discretization scheme for ϕ_e and ϕ_w
(Interpolation of transport property to cell face)

Central differencing scheme

- Substitute in transport equation:

$$\phi_e = (\phi_P + \phi_E) / 2$$

$$\phi_w = (\phi_W + \phi_P) / 2$$



- Rearrangement leads to CDS:

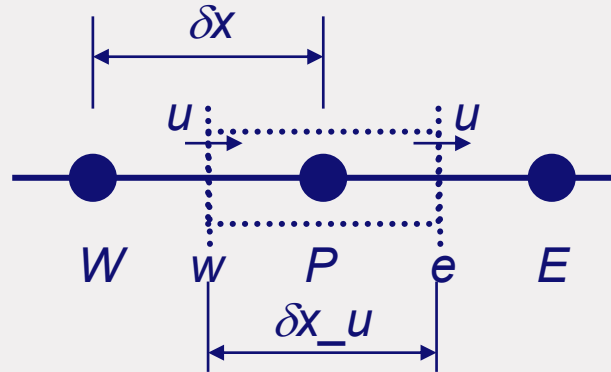
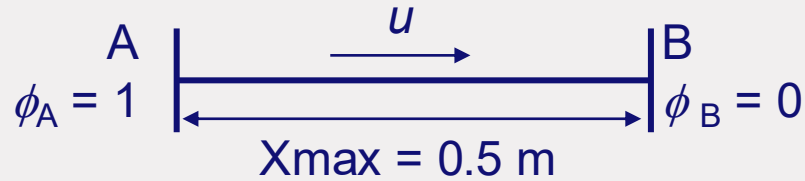
$$F_e \phi_e - F_w \phi_w = D_e (\phi_E - \phi_P) - D_w (\phi_P - \phi_W) - D_w (\phi_P - \phi_W)$$

$$a_P \phi_P = a_W \phi_W + a_E \phi_E$$

a_W	a_E	a_P
$D_w + \frac{F_w}{2}$	$D_e - \frac{F_e}{2}$	$a_W + a_E + (F_e - F_w)$

Solved for all grid nodes to obtain the distribution of ϕ

1D convection-diffusion of ϕ



Numerical Solution

```
%% CONSTANTS      /* define all the constants */
%% INIT           /* initialize variables */
%% BOUNDARY       /* apply boundary conditions */
%% phi-EQUATION   /* calculate coefficients aE, aW, aP, b */
%% SOLVE /* solver iteration loop */
    for iter = 0:OUTER_ITER
        ... /* solve1D(T, b, aE, aW, aP) */
        fprintf(...); /* write convergence to screen */
    end
%% OUTPUT         /* write output to file */
```

Calculation of coefficients a_{nb} , a_P , b

for I = Istart:Iend

i = I;

Fw = 0.5*(rho(I-1) + rho(I))*u(i); */* convective flux */*

Fe = 0.5*(rho(I+1) + rho(I))*u(i+1); */* convective flux */*

Dw = 0.5*(Gamma(I-1) + Gamma(I))/(x(I) - x(I-1)); */* diffusion */*

De = 0.5*(Gamma(I+1) + Gamma(I))/(x(I+1) - x(I)); */* diffusion */*

SP(I) = 0.; */* coefficient of the linearised source term */*

Su(I) = 0.; */* constant part of the source term */*

aW(I) = Dw + 0.5*Fw; */* coefficient for TW using CDS */*

aE(I) = De - 0.5*Fe; */* coefficient for TE using CDS */*

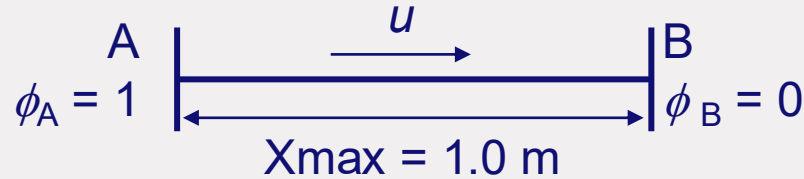
aP(I) = aW(I) + aE(I) + Fe - Fw - SP(I); */* coefficient for TP */*

b(I) = Su(I); */* constant term in the discretisation equation */*

end

a_W	a_E	a_P
$D_w + \frac{F_w}{2}$	$D_e - \frac{F_e}{2}$	$a_W + a_E + (F_e - F_w)$

Example 1: low speed, CDS

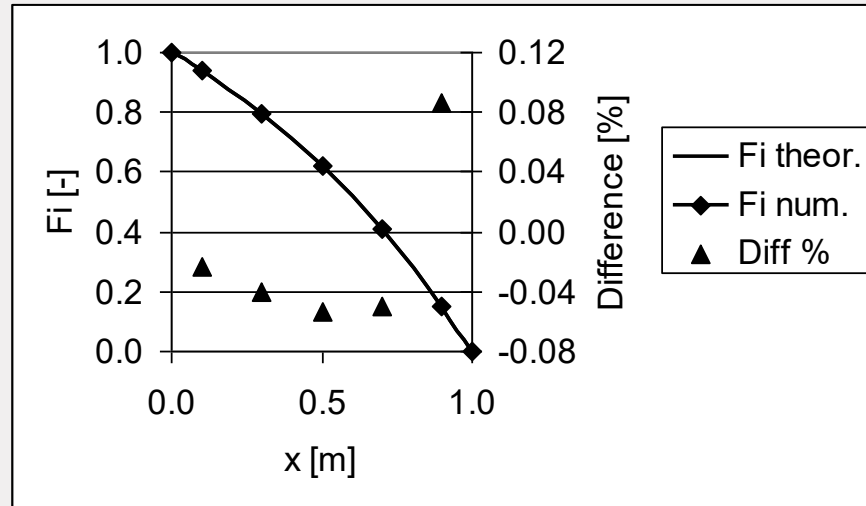


$$u = 0.1 \text{ m/s}, NPI = 5$$

$$F = \rho u = 0.1$$

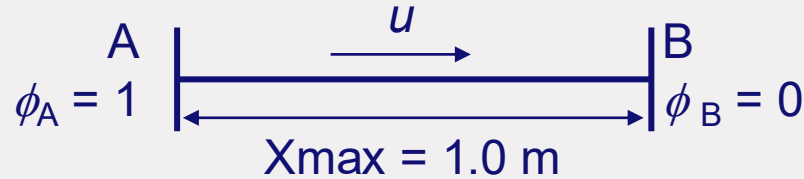
$$D = \Gamma / \delta x = 0.1 / 0.2 = 0.5$$

$$Pe = F / D = 0.2$$



Good predictions with
only 5 grid points!

Example 2: high speed, CDS

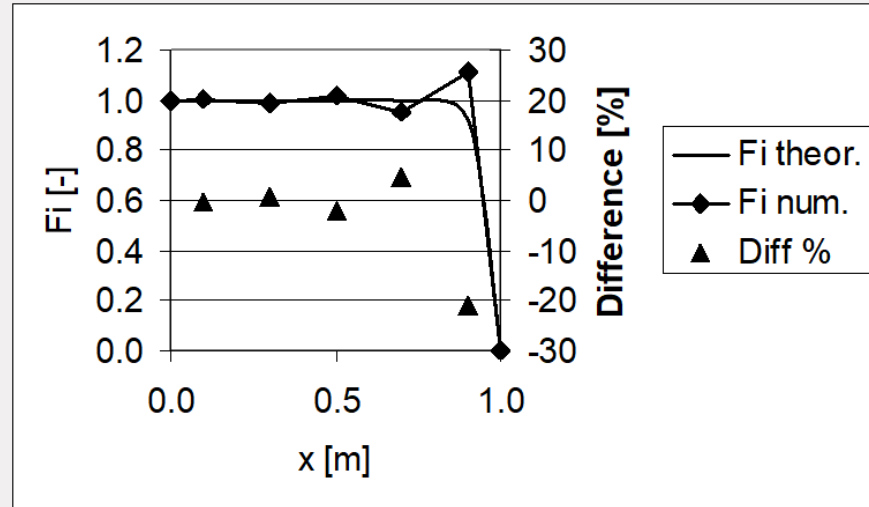


$$u = 2.5 \text{ m/s}, NPI = 5$$

$$F = \rho u = 2.5$$

$$D = \Gamma / \delta x = 0.1 / 0.2 = 0.5$$

$$Pe = F / D = 5.0$$



Reasonable predictions.
CDS gives **overshoots!**

Properties of discretization schemes

1. Conservative

2. Boundedness

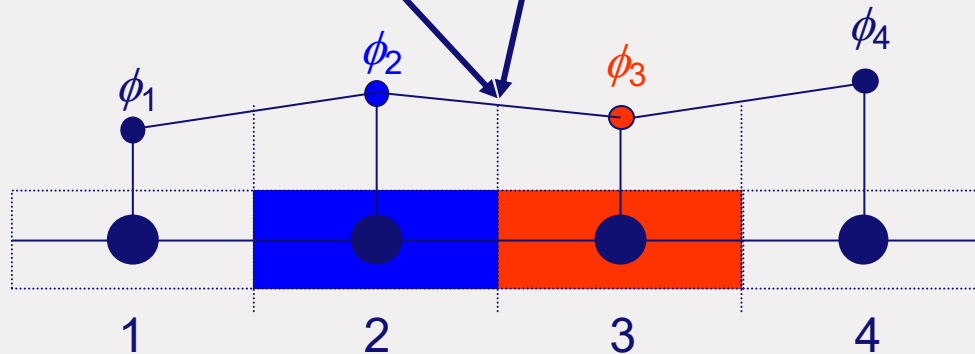
3. Transportiveness

Properties discretization schemes

1. Conservative:

Consistent treatment of fluxes through nodal faces

$$\text{flux} = \left(\Gamma \frac{d\phi}{dx} \right)_e = \left(\Gamma \frac{d\phi}{dx} \right)_w = \Gamma \frac{(\phi_3 - \phi_2)}{\delta x} \quad (\text{Same expression!})$$



Properties discretization schemes

2. Boundedness:

- property ϕ_p is bounded by values of ϕ at the boundaries, i.e. no overshoots or undershoots:

$$\frac{\sum_{nb} |a_{nb}|}{|a_p - S_p|} \begin{cases} \leq 1 \text{ at all nodes} \\ < 1 \text{ at one node at least} \end{cases}$$

- we need to satisfy:

$$S_p < 0$$

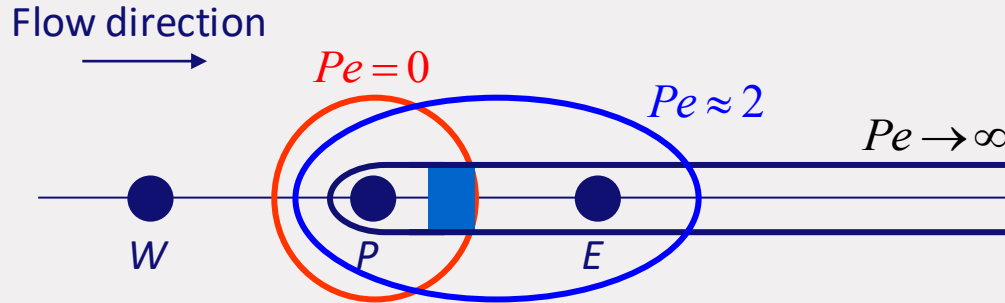
$$a_p, a_{nb} > 0$$

Properties discretization schemes

3. Transportiveness:

- Direction of influence = $f(Pe)$
- No convection and pure diffusion ($Pe = 0$)
- No diffusion and pure convection ($Pe \rightarrow \infty$)

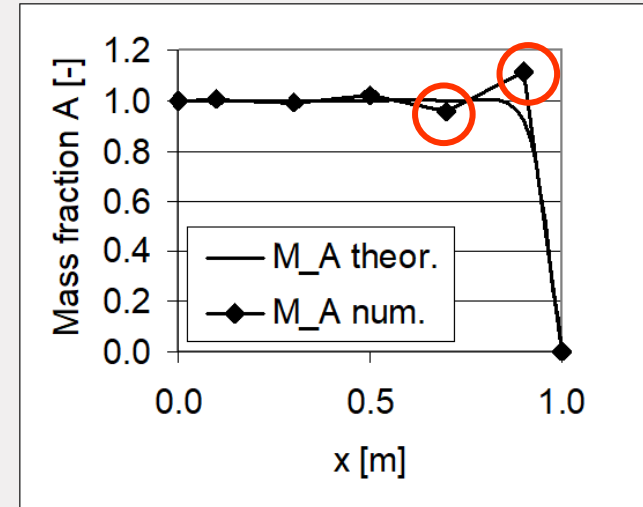
$$Pe = F / D = \frac{\rho u}{\Gamma / \delta x}$$



Area of influence of variable ϕ_p as function of Pe

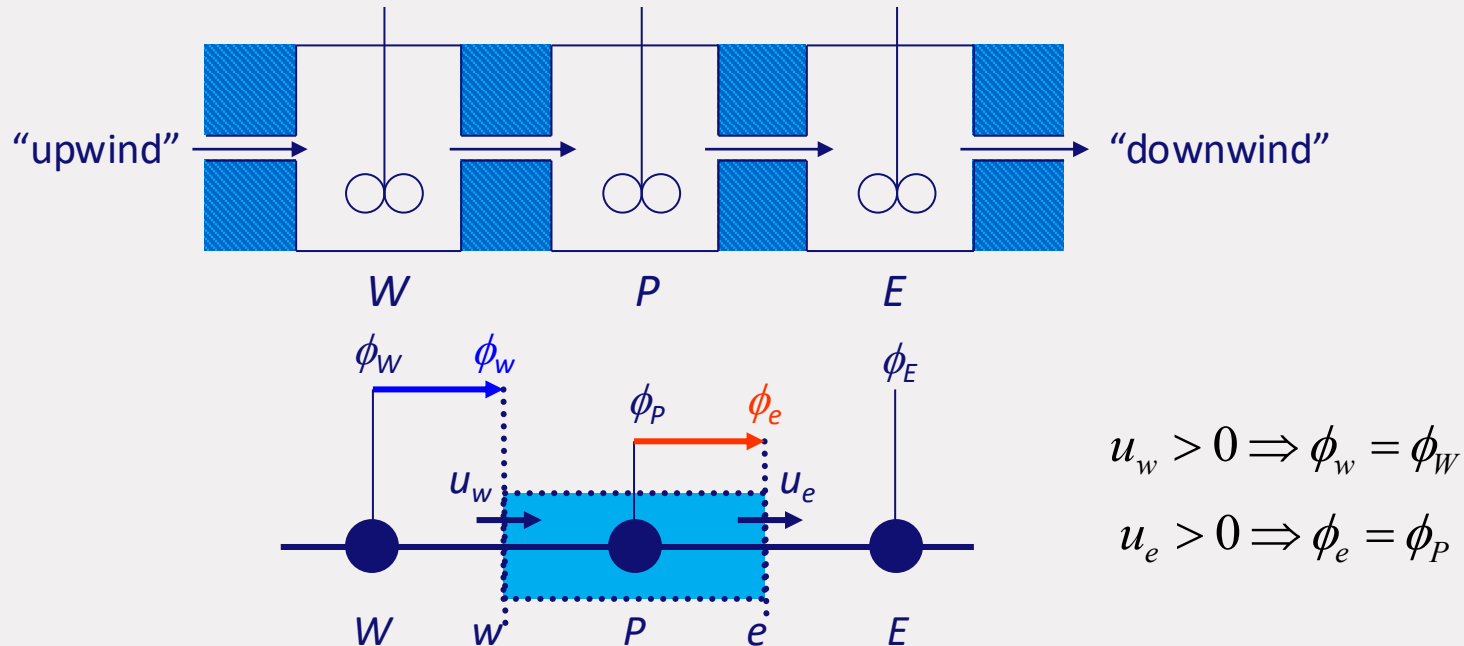
Assessment CDS

- **Conservativeness satisfied**
- **Boundedness not always satisfied**
 - Example 2: $a_E = D_e - \frac{F_e}{2} = 0.5 - 1.25 < 0!$
 - CDS only satisfactory if $a_E > 0 \Rightarrow F_e / D_e = Pe_e < 2$
- **Transportiveness not always satisfied**
 - Example 1: $Pe = 0.2 \rightarrow$ CDS works well
 - Example 2: $Pe = 5.0 \rightarrow$ CDS gives overshoots
 - CDS scheme does not recognize flow direction!



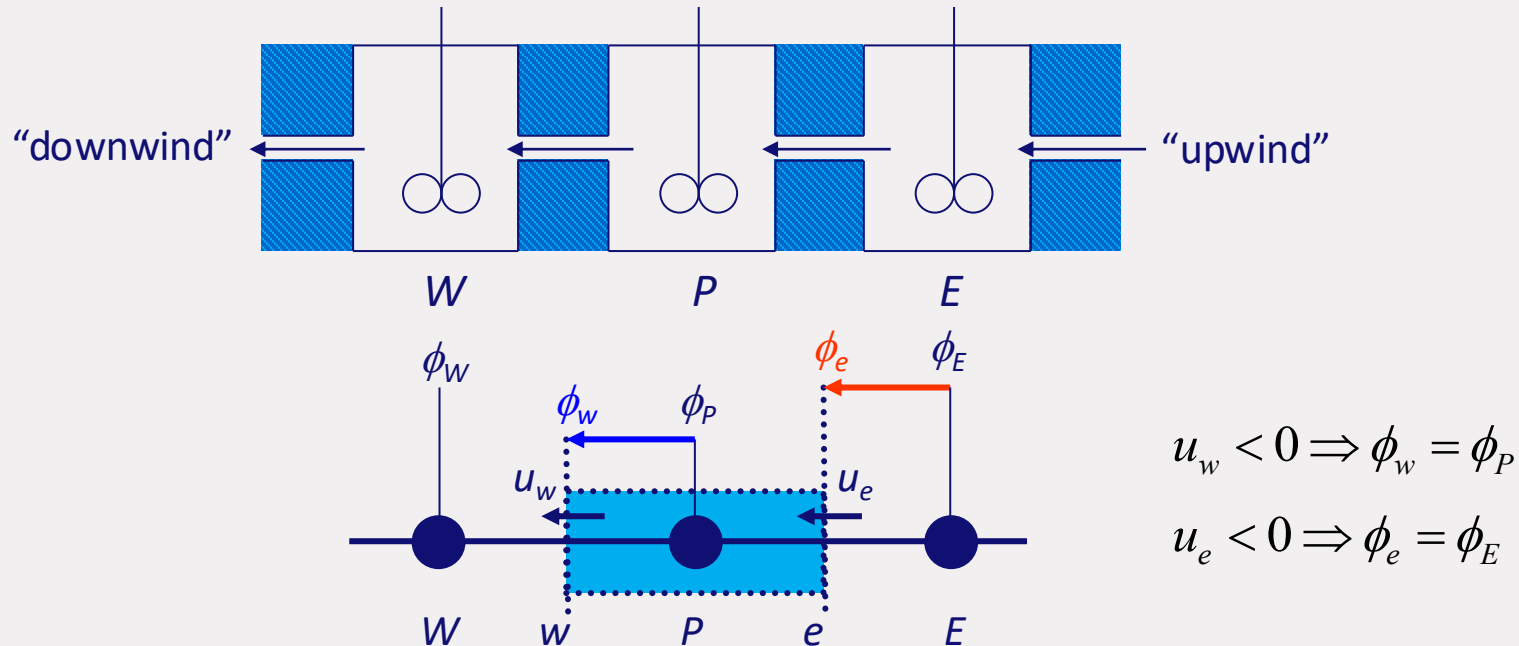
Upwind differencing scheme

UDS resembles CISTRs (Continuous Ideally Stirred-Tank Reactor) in series



Upwind differencing scheme

UDS resembles CISTRs in series



Upwind differencing scheme

- Integrated convection-diffusion equation:

$$(\rho u A \phi)_e - (\rho u A \phi)_w = \left(\Gamma A \frac{d\phi}{dx} \right)_e - \left(\Gamma A \frac{d\phi}{dx} \right)_w$$

$$F_e \phi_e - F_w \phi_w = D_e (\phi_E - \phi_P) - D_w (\phi_P - \phi_W)$$

- Substitute ϕ_e and ϕ_w in transport equation:

$$u_w > 0, u_e > 0 \Rightarrow F_w > 0, F_e > 0 \Rightarrow \phi_w = \phi_W, \phi_e = \phi_P$$

$$F_e \phi_P - F_w \phi_W = D_e (\phi_E - \phi_P) - D_w (\phi_P - \phi_W)$$

$$u_w < 0, u_e < 0 \Rightarrow F_w < 0, F_e < 0 \Rightarrow \phi_w = \phi_P, \phi_e = \phi_E$$

$$F_e \phi_E - F_w \phi_P = D_e (\phi_E - \phi_P) - D_w (\phi_P - \phi_W)$$

Upwind differencing scheme

- Rewrite convection-diffusion equation in general form:

$$a_P \phi_P = a_W \phi_W + a_E \phi_E + S_u$$

- Coefficients for all u ($u < 0$ and $u > 0$):

a_W	a_E	a_P
$D_w + \max(F_w, 0)$	$D_e + \max(-F_e, 0)$	$a_W + a_E + (F_e - F_w) - S_P$

Calculation of coefficients a_{nb} , a_p , b

for I = Istart:Iend

```

i = I;
Fw = 0.5*(rho(I-1) + rho(I))*u(i); /* convective flux */
Fe = 0.5*(rho(I+1) + rho(I))*u(i+1); /* convective flux */
Dw = 0.5*(Gamma(I-1) + Gamma(I))/(x(I) - x(I-1)); /* diffusion */
De = 0.5*(Gamma(I+1) + Gamma(I))/(x(I+1) - x(I)); /* diffusion */

SP(I) = 0.; /* coefficient of the linearised source term */
Su(I) = 0.; /* constant part of the source term */

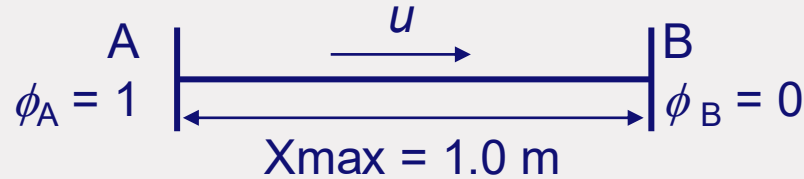
aW(I) = Dw + max( Fw, 0.); /* coefficient for TW using UDS */
aE(I) = De + max(-Fe, 0.); /* coefficient for TE using UDS */
aP(I) = aW(I) + aE(I) + Fe - Fw - SP(I); /* coefficient for TP */
b(I) = Su(I); /* constant term in the discretisation equation */

```

end

a_w	a_E	a_p
$D_w + \max(F_w, 0)$	$D_e + \max(-F_e, 0)$	$a_w + a_E + (F_e - F_w) - S_P$

Example 1: low speed, UDS

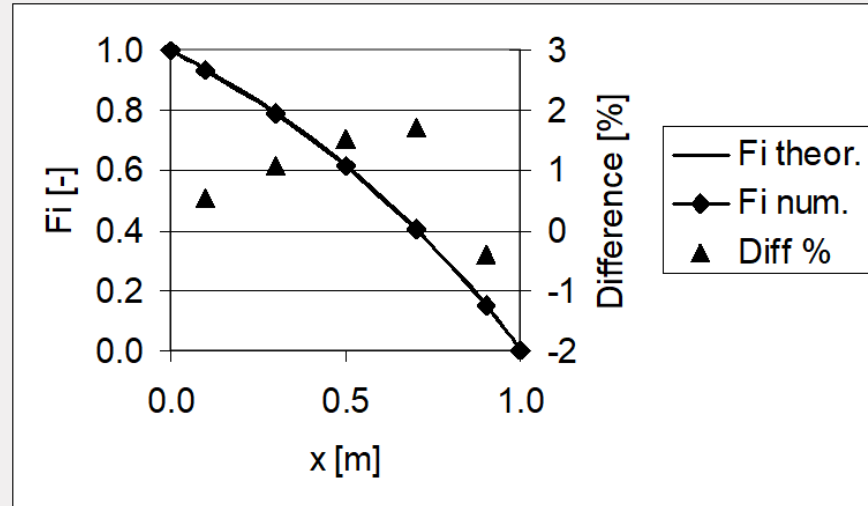


$$u = 0.1 \text{ m/s}, NPI = 5$$

$$F = \rho u = 0.1$$

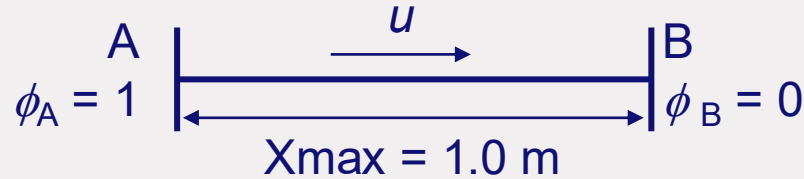
$$D = \Gamma / \delta x = 0.1 / 0.2 = 0.5$$

$$Pe = F / D = 0.2$$



Good predictions with
only 5 grid points!

Example 2: high speed, UDS

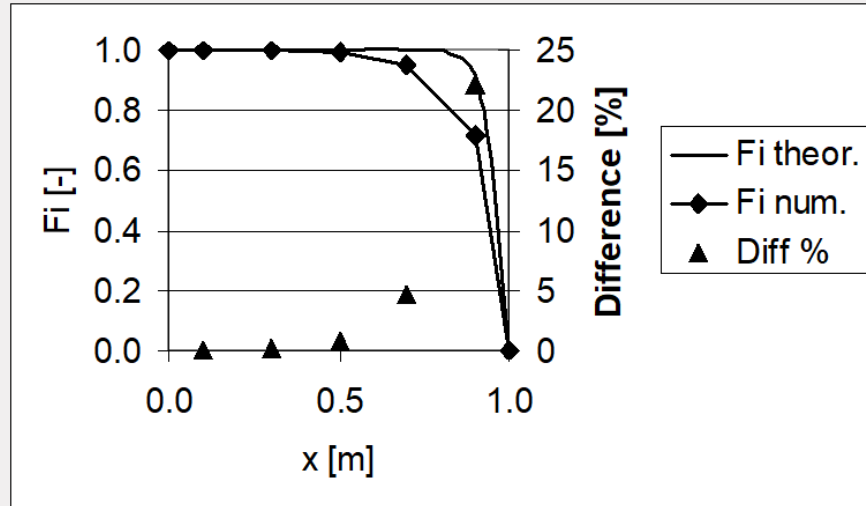


$$u = 2.5 \text{ m/s}, NPI = 5$$

$$F = \rho u = 2.5$$

$$D = \Gamma / \delta x = 0.1 / 0.2 = 0.5$$

$$Pe = F / D = 5.0$$

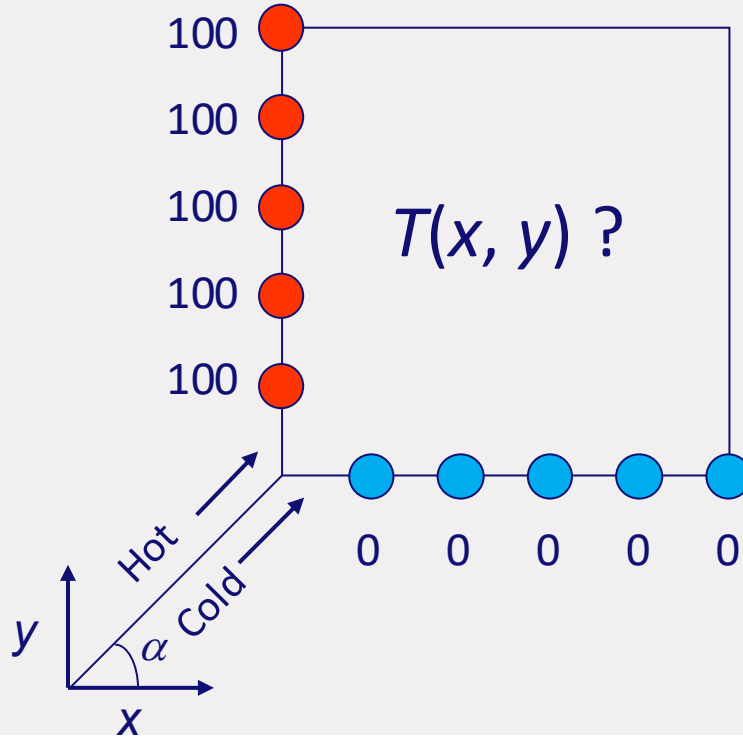


Good predictions.
UDS gives no overshoots.

Assessment Upwind scheme

- **Conservativeness satisfied**
- **Boundedness satisfied**
 - All coefficients $a_{nb} > 0$
 - No restrictions on Pe
- **Transportiveness satisfied**
 - UDS scheme accounts for flow direction

Example 2D pure convection

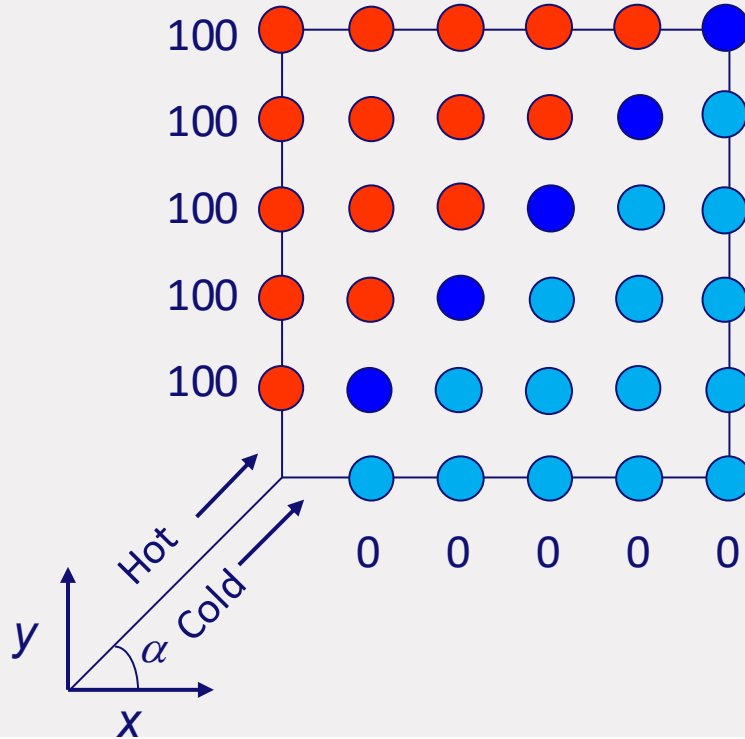


$$F = \rho u = 1$$

$$\Gamma = 0 \Rightarrow D = \Gamma / \delta x = 0$$

$$Pe = F / D \rightarrow \infty$$

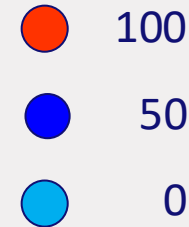
Example 2D pure convection



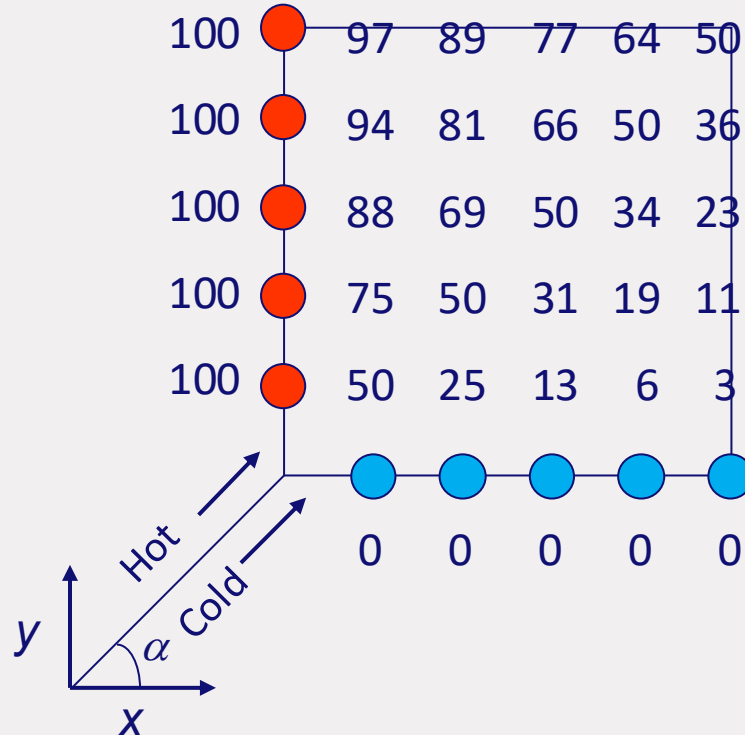
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$$Pe = F / D \rightarrow \infty$$



Example 2D pure convection



$$F = \rho u = 1$$

$$\Gamma = 0 \Rightarrow D = \Gamma / \delta x = 0$$

$$Pe = F / D \rightarrow \infty$$

“false diffusion” error

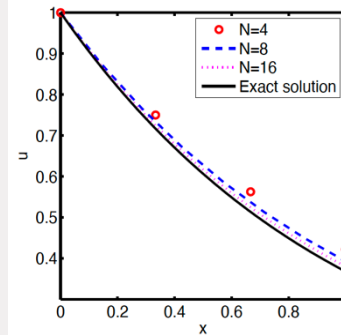
Accuracy Upwind scheme

- UDS $\equiv$ 1D interpolation
- In 2D or 3D when the flow is not aligned with the grid lines, UDS gives “false diffusion”
- False diffusion only occurs at high Pe
- Minimize false diffusion by:
 - Aligning grid with flow direction
 - Using fine grids
 - Using more accurate differencing schemes

Spatial discretization error:

$$\text{e.g., } \left(\frac{\partial \phi}{\partial x} \right)_P = \frac{\phi_E - \phi_P}{\Delta x} - \left(\frac{\partial^2 \phi}{\partial x^2} \right)_P \frac{\Delta x}{2} - \dots$$

Spatial convergence, grid (in)dependence, grid convergence, grid refinement

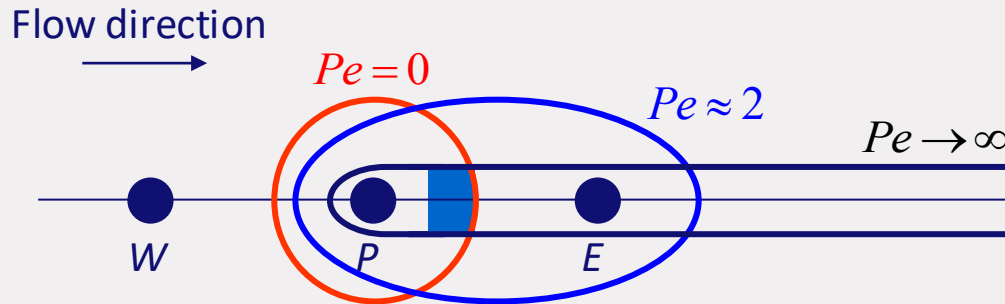


Practice in tutorial

Hybrid differencing scheme

Hybrid scheme combines both schemes:

- CDS is $O(2)$ accurate for $Pe < 2$
- UDS is $O(1)$ accurate and accounts for flow direction for $Pe \geq 2$



Area of influence of variable ϕ_p as function of Pe

Hybrid differencing scheme

- If $Pe < 2$, use CDS
- If $Pe \geq 2$, use UDS without diffusion

	a_w	a_E
CDS	$D_w + F_w / 2$	$D_e - F_e / 2$
UDS	$D_w + \max(F_w, 0)$	$D_e + \max(-F_e, 0)$
Hybrid	$\max(F_w, (D_w + F_w / 2), 0)$	$\max(-F_e, (D_e - F_e / 2), 0)$

Assessment Hybrid scheme

- **Hybrid scheme combines CDS and UDS**
 - **Conservativeness satisfied**
 - **Boundedness satisfied**
 - ✓ All coefficients $a_{nb} > 0$
 - ✓ No restrictions on Pe
 - **Transportiveness satisfied**
 - ✓ Scheme accounts for flow direction
- **Advantages: robust and reliable**
- **Disadvantage: only $O(1)$ accuracy**

Other schemes

- **QUICK $O(3)$, higher order Upwind $O(2)$**
 - Uses extra grid points $\phi_{WW}, \phi_W, \phi_P, \phi_E, \phi_{EE}$
 - Higher accuracy at fine grids
 - Boundedness not always guaranteed: overshoots
- **Total variation diminishing (TVD) scheme**
 - Van Leer, Superbee, Min-Mod, MUSCL, etc.
 - $O(2)$ accuracy, boundedness satisfied
- **See also book Versteeg and Malasekera**

Wrap up

- **convection** = **diffusion** + **(source – sink)**

$$\text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- Differencing schemes needed for ϕ_e and ϕ_w
- **General purpose schemes: UDS and Hybrid**
Disadvantage: false or numerical diffusion in multi-dimensional flows
- **General form of discretized equations:**

$$a_P\phi_P = \sum a_{nb}\phi_{nb} + S_u$$

$$a_P = \sum a_{nb} + \Delta F - S_P$$